Bayesian data analysis

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Overview

Today, we cover:

- Revisiting Gaussian Quadrature
- Bayesian data analysis crash course
- Motivation for MCMC

Tomorrow:

- MCMC

Announcements:

- Cloud computing (Keven Haynes) moved to March 26
    - Should be getting emails from Keven, please respond
- We will finish up Bayesian computing after break

Gaussian quadrature

Gaussian quadrature allows subintervals to be more freely chosen.

- Weights and subintervals are node-specific
- Provides the **exact solution** if f is a polynomial with degree at most $2n - 1$

$$\int_a^b \omega(x) f(x) dx \approx \sum_{j=1}^{n} w_j f(x_j)$$

- $\omega(x)$ is some weight function
- $w_j, j = 1, \dots, n$: are specific weights; x_j are nodes
- If f can be approximated well by a degree $2n - 1$ polynomial on $[a, b]$, then Gaussian quadrature provides a good approximation to the integral of interest.

Gaussian quadrature

Question: How to choose the weights and placement of the nodes?

- Gauss-Legendre quadrature: $\int_{-1}^{1} f(x)dx$
- Gauss-Chebyshev: $\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} f(x)dx$
- Gauss-Laguerre: $\int_{0}^{\infty} e^{-x} f(x)dx$
- Gauss-Hermite: $\int_{-\infty}^{\infty} e^{-x^2} f(x)dx$

Two-point Gauss-Legendre example

Two-point Gauss-Legendre example

General Gauss-Legendre

The weights for Gauss-Legendre quadrature are given by:

$$w_i = \frac{2}{(1 - x_i^2)[P_n'(x_i)]^2}$$

Where the first few Legendre polynomials are:

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2}\left(3x^2 - 1\right)$$

$$P_3(x) = \frac{1}{2}\left(5x^3 - 3x\right)$$

$$P_4(x) = \frac{1}{8}\left(35x^4 - 30x^2 + 3\right)$$

$$P_5(x) = \frac{1}{8}\left(63x^5 - 70x^3 + 15x\right)$$

Gauss-Legendre nodes and weights

Number of points, n	Points, x_i		Weights, w_i	
1	0		2	
2	$\pm\frac{1}{\sqrt{3}}$	±0.57735...	1	
3	0		$\frac{8}{9}$	0.888889...
	$\pm\sqrt{\frac{3}{5}}$	±0.774597...	$\frac{5}{9}$	0.555556...
4	$\pm\sqrt{\frac{3}{7}-\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.339981...	$\frac{18+\sqrt{30}}{36}$	0.652145...
	$\pm\sqrt{\frac{3}{7}+\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.861136...	$\frac{18-\sqrt{30}}{36}$	0.347855...
5	0		$\frac{128}{225}$	0.568889...
	$\pm\frac{1}{3}\sqrt{5-2\sqrt{\frac{10}{7}}}$	±0.538469...	$\frac{322+13\sqrt{70}}{900}$	0.478629...
	$\pm\frac{1}{3}\sqrt{5+2\sqrt{\frac{10}{7}}}$	±0.90618...	$\frac{322-13\sqrt{70}}{900}$	0.236927...

The Bayesisan philosophy : a simple motivating example

Imagine we are interested in the daily number of steps taken by Rollins students. We use accelerometers to collect a random sample of step counts from $n = 10$ students.

- Let y_i represent the step count for the ith student
- Assume $y_i \sim N(\mu, \sigma_y^2)$

We want to learn about μ, the average daily number of steps taken by Rollins students.

The frequentist approach

We want to learn about μ, the average daily number of steps taken by Rollins students.

- Parameter μ is fixed and unknown
- The data y is random
- The sample mean $\hat{\mu} = \bar{y}$ is a statistic, and a frequentist estimator of the population mean μ.
- Base inference on p-values and confidence intervals

Frequentist inference about μ

```r
head(steps)
```

```
[1]  9732  9270 10446 10699 10336 10326
```

```r
mean(steps)
```

```
[1] 9999.2
```

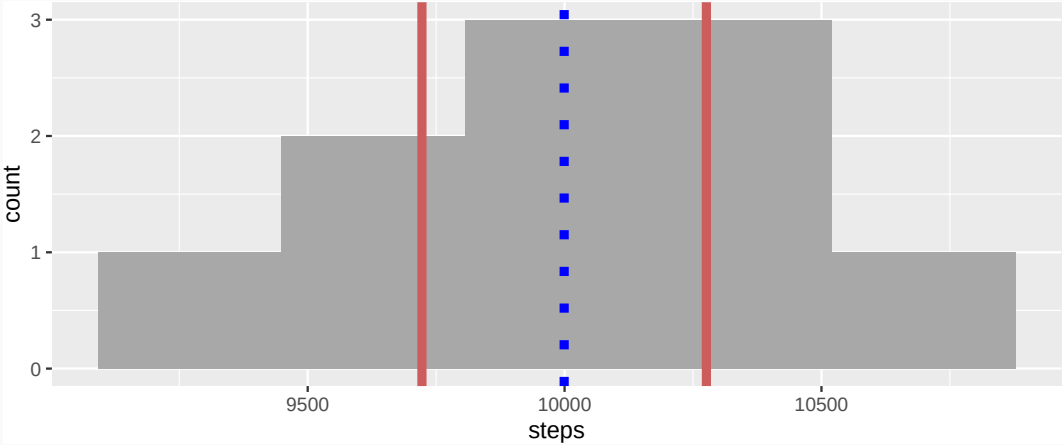
```r
ci_minus = ybar - 1.96 * sd(steps) / sqrt(n)
ci_plus = ybar + 1.96 * sd(steps) / sqrt(n)
ci_minus
```

```
[1] 9722.341
```

```r
ci_plus
```

```
[1] 10276.06
```

Frequentist inference about μ



The Bayesian approach

- Bayesians treat the parameter μ as random
- Express uncertainty about μ using probability distributions

- The distribution before observing the data is called the **prior distribution**
  - Allows incorporation of previous knowledge

- Distribution after observing the data is the **posterior distribution**
- Inference is conditional on the data we observe

The Bayesian approach

What do I think I know?

- $y_i|\mu \sim N(\mu, \sigma_y^2)$
- $\mu \sim N(\mu_0, \sigma_\mu^2)$

What do I want to learn?

- $\mu|y_i \sim$???

This is a **one sample Normal-Normal** model

The Bayesian approach

Relate prior, likelihood, and posterior through Bayes' formula:

$$p(\mu \mid y_i) = \frac{p(y_i \mid \mu)\, p(\mu)}{p(y_i)} \tag{1}$$

$$= \frac{p(y_i \mid \mu)\, p(\mu)}{\int_\mu p(y_i \mid \mu)\, p(\mu)\, d\mu} \tag{2}$$

$$\propto p(y_i \mid \mu)\, p(\mu) \tag{3}$$

For the Normal likelihood with a Normal prior for μ, the posterior is also Normal:

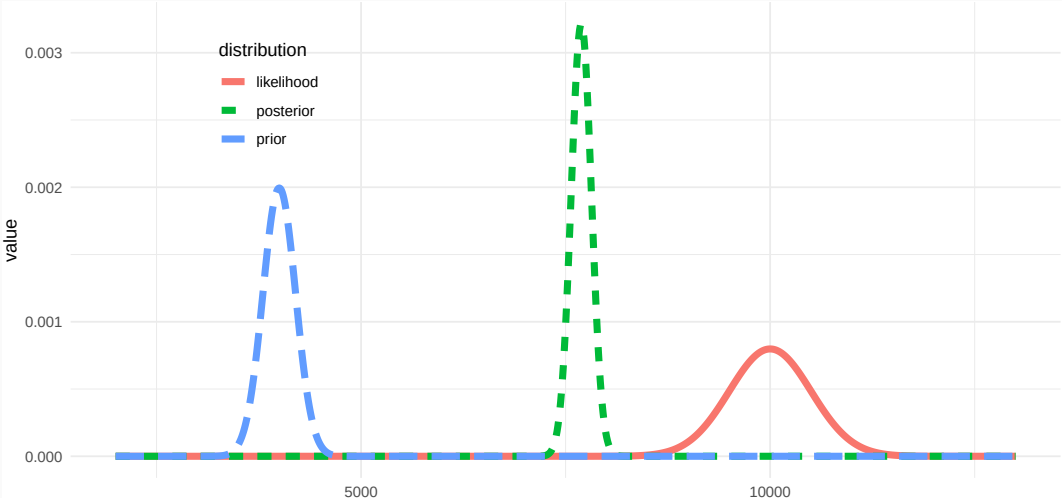
$$\mu | y_i \sim N\left(\frac{\sigma_\mu^2}{\sigma_y^2/n + \sigma_\mu^2} \bar{y} + \frac{\sigma_y^2/n}{\sigma_y^2/n + \sigma_\mu^2} \mu_0, \frac{\sigma_y^2 \sigma_\mu^2/n}{\sigma_y^2/n + \sigma_\mu^2} \right)$$

Choosing the prior distribution

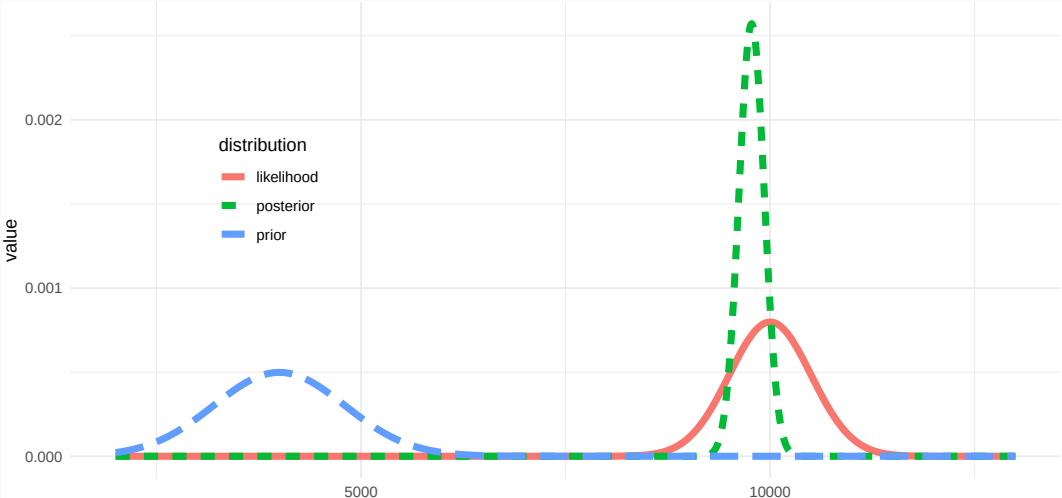
A study in *Medicine & Science in Sports & Exercise* reports that Americans take an average 4000 steps per day with a standard deviation of 200 steps. We come up with three different priors:

- **Informative prior**: Maybe we really believe the previous study
  - $\mu \sim N(4000, 200^2)$
- **Weakly informative prior**: Or maybe we believe it a little bit
  - $\mu \sim N(4000, 800^2)$
- **Uninformative or diffuse prior**: Maybe we don't have any scientific information at all
  - $\mu \sim N(0, 5000^2)$

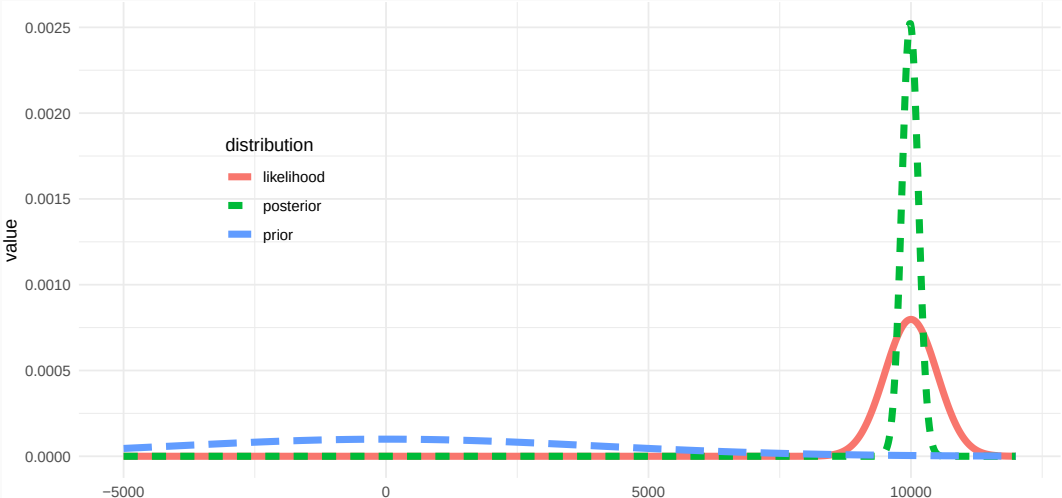
Effect of informative prior



Effect of weakly informative prior



Effect of uninformative prior



Bayesian inference about μ

We decide to go with the uninformative prior.

- Posterior mean: $\mu|y = 9984$ steps
- Posterior 95% **credible interval** is the (2.5%, 97.5%) quantile of the posterior distribution

```r
qnorm(c(0.025, 0.927), mu_post, sigma_post)
```

```
[1]  9673.945 10213.288
```

A **Bayesian credible interval** for a parameter θ is an interval $[a, b]$ s.t.

$$P(\theta \in [a, b]|y) = 1 - \alpha$$

- Given the data, there is a 95% probability that the true number of steps for Rollins students lies within $[9673, 10213]$

Bayesian inference about μ

- Posterior probability that $\mu < 10000$ given y

```r
pnorm(10000, mu_post, sigma_post)
```

```
[1] 0.5413359
```

Bayesian inference allows us to make probability statements about μ given the data.

Linear regression- frequentist

Frequentist approach: estimate β and σ^2

- $\hat{\beta} = (X^T X)^{-1} X^T y$
- Compute residuals using $\hat{\beta}$, then from residuals estimate σ^2

Steps for a Bayesian linear regression

- Assign priors to the parameters of interest
  - Normal prior for β
  - Inverse gamma for σ^2
- Choose hyper-parameters
  - Prior mean and variance for β
  - Shape and scale for σ^2
- Obtain joint posterior distribution, and base inference on this

Bayesian linear regression (known variance)

We want a Bayesian framework for the regression model

$$y = X\beta + \epsilon; \epsilon \sim N(0, \sigma_y^2 I_n)$$

- Need to make distributional assumptions about β
  - $\beta \sim N(0, \sigma_b^2 I_p)$

Bayesian regression (known variance)

We want to obtain the posterior

$$p(\beta|y, X) \propto p(y|\beta, X)p(\beta)$$

Bayesian regression (known variance)

Bayesian regression (known variance)

$$\beta \mid y \sim \mathcal{N}(\beta_{\text{post}}, \Sigma_{\text{post}})$$

$$\Sigma_{\text{post}} = \left(\frac{1}{\sigma_y^2} X^\top X + \frac{1}{\sigma_\beta^2} I_p \right)^{-1}$$

$$\beta_{\text{post}} = \Sigma_{\text{post}} \left(\frac{1}{\sigma_y^2} X^\top y + \frac{1}{\sigma_\beta^2} \beta_0 \right) \tag{4}$$

$$= \left(X^\top X + \frac{\sigma_y^2}{\sigma_\beta^2} I_p \right)^{-1} (X^\top y) \tag{5}$$

About the variances

Throughout all of this we have implicitly conditioned on the variances σ_y^2 and σ_β^2

How do we "choose" the variance terms?

Prior variance

The prior variance σ_β^2 is often pre-selected to indicate the amount of prior knowledge

- Typically this is chosen very large to indicate a lack of knowledge
- This is considered a hyperparameter
- Basically this means that your prior will be dominated by the data and likelihood

Outcome variance

The outcome variance σ_y^2 is a quantity of interest.

- Ideally, we'd like to estimate this using the observed data
- Since we're already being Bayesians, why not assign a prior distribution and try to find a posterior?

The inverse-gamma distribution works pretty well

$$p(\sigma_y^2|A, B) = \frac{B^A}{\Gamma(A)}(\sigma_y^2)^{-A-1}\exp(-\frac{B}{\sigma_y^2})$$

- Support of σ_y^2:

Outcome variance posterior

We want to find

$$p(\sigma_y^2|y, X, \beta) \propto p(y|\beta, \sigma_y^2, X)p(\sigma_y^2|A, B)$$

It turns out that the posterior outcome variance is also inverse Gamma... this is an example of a **conjugate prior**.

Some Bayesian language

- Parameters of interest are β, σ_y^2
- The priors we've used are called **conjugate priors**
- The hyperparameters we've chosen are σ_b^2, A, B
- The distributions we've calculated are $p(\sigma_y^2|y, X, \beta)$ and $p(\beta|y, X, \sigma_y^2)$
  - These are called **full conditionals**
- The posterior distribution of interest is $p(\beta, \sigma_y^2|y, X)$, which is called the **joint posterior**

Conjugate priors

A prior is **conjugate** if the posterior is a member of the same parametric family. Some examples are:

- **beta-binomial** model: if the response is binomial and we use a beta prior, the posterior is beta
- **gamma-Poisson**: Poisson reponse + gamma prior = gamma posterior
- **normal-normal** model: normal response + normal prior = normal posterior

Advantage of a conjugate prior is that the posterior is available in closed form

Back to the joint posterior distribution

Our goal is to learn about the joint posterior distribution

$$p(\beta, \sigma_y^2 | y, X, hyperparameters)$$

We can calculate this from the full conditionals using the law of conditional probability:

$$p(\beta, \sigma_y^2 | y, X, h) = p(\beta | \sigma_y^2, y, X, h) p(\sigma_y^2, X, h) \tag{6}$$

$$= p(\beta | \sigma_y^2, y, X, h) \int_\beta p(\sigma_y^2 | \beta, y, X, h) d\beta \tag{7}$$

Joint posterior distribution

Our goal is to learn about the joint posterior distribution

$$p(\beta, \sigma_y^2 | y, X, hyperparameters)$$

This can be hard!

- Often the joint posterior is analytically intractable
- For the conjugate priors we use, though, the full conditionals were "easy"
- If we can't write down the joint posterior, maybe we can sample from it

MCMC

Basic idea: How do I get a string of values from the posterior when I cannot write the posterior down?

- Markov chain Monte Carlo (MCMC) methods let you sample from complicated [posterior] distributions
  - a collection of tools used to sample from a target distribution
- Once you have a sample from the posterior distribution you can use summary statistics (mean, variance, quantiles) to do posterior inference

There are lots of MCMC methods-

- We're going to talk about a couple of the simpler ones.
  - Metropolis-Hastings
  - Gibbs sampler

Steps for a Bayesian regression

- Assign priors to the parameters of interest β, σ_y^2
- Choose hyper-parameters
  - Prior mean and variance for β
  - Distributional parameters for σ_y^2
- Obtain joint posterior distribution, and base inference on this

We're going to do this using data from a cannabis consumption and driving study in Colorado.

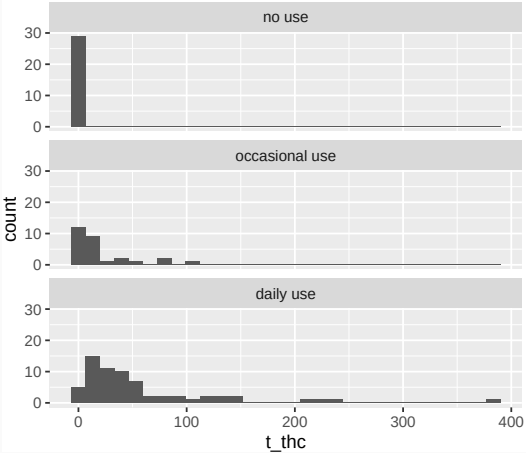
Cannabis data

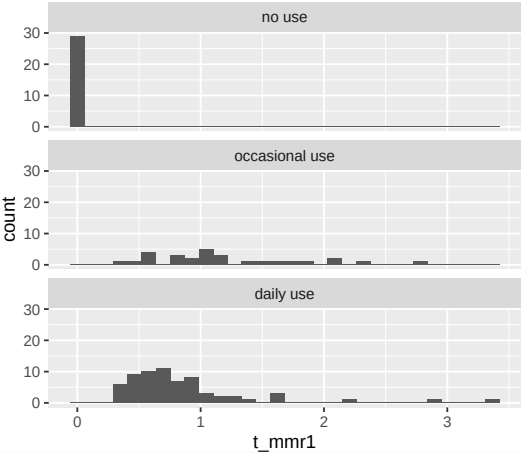
```
cannabis = readRDS(file = here::here("data", "cannabis.rds"))
cannabis = cannabis %>% select(id, use_group, t_mmr1, t_thc, p_fpc1, p_change = p_PMC_pctCh
                               i_memory_time34, i_time_outside_reticle) %>%
  mutate(log_mmr = log1p(t_mmr1))


head(cannabis)
```

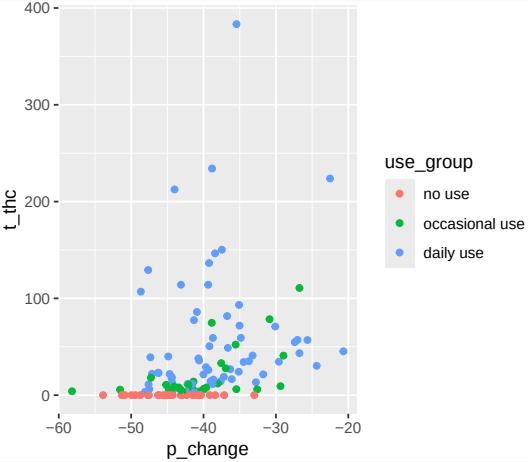
```
       id use_group t_mmr1    t_thc     p_fpc1 p_change i_memory_time34
1 003-001 daily use 0.6108 146.5474  38.987583 -38.4050        3.096037
2 003-003 daily use 0.5093  57.0107 161.109915 -27.0250        3.591851
3 003-004 daily use 1.5778  22.0922 -77.269994 -47.1130        4.523515
4 003-005 daily use 0.6864  81.7211  37.433740 -36.7350        4.338389
5 003-006 daily use 0.4474  14.3114  58.375202 -37.6325        2.802600
6 003-007 daily use 0.5221   9.9674   5.019542 -44.9335        4.706120
  i_time_outside_reticle   log_mmr
1                70.9271 0.4767309
2                66.1071 0.4116460
```

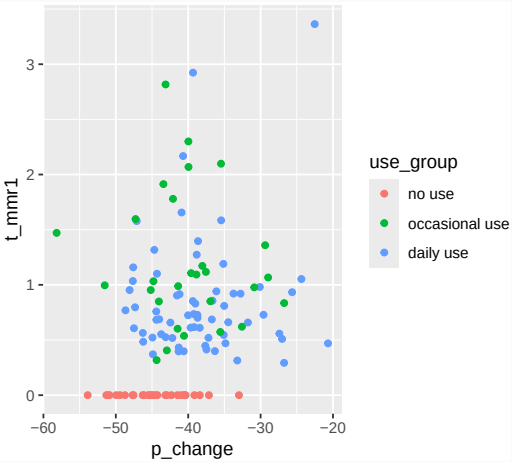
Cannabis data



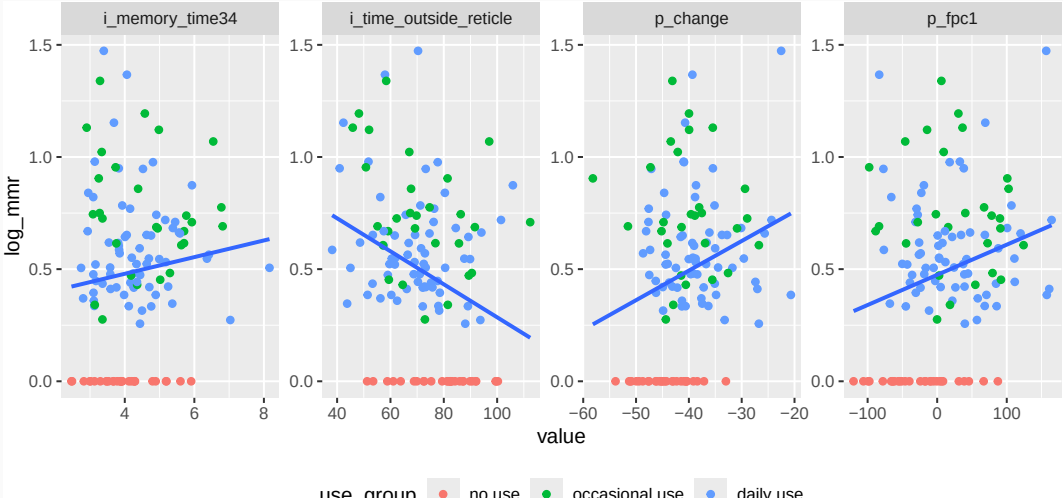


Cannabis data

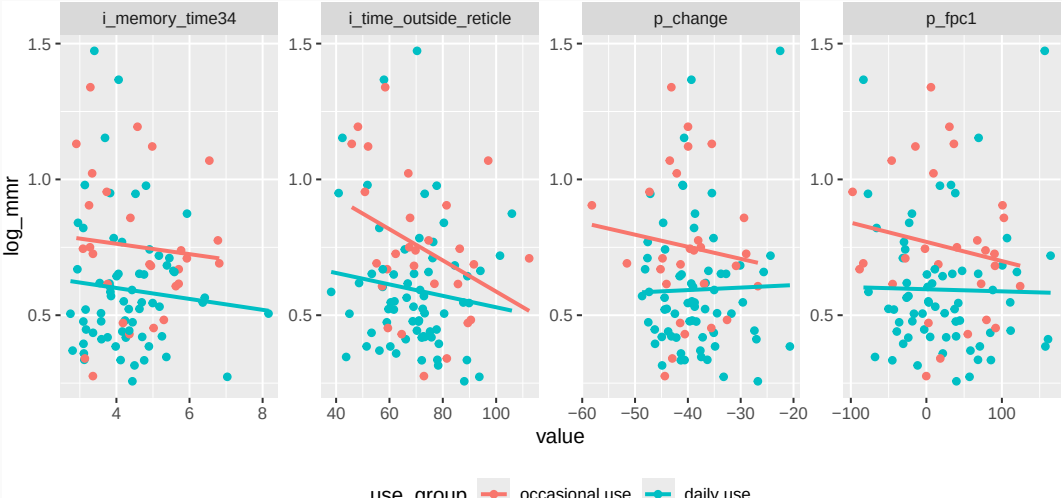




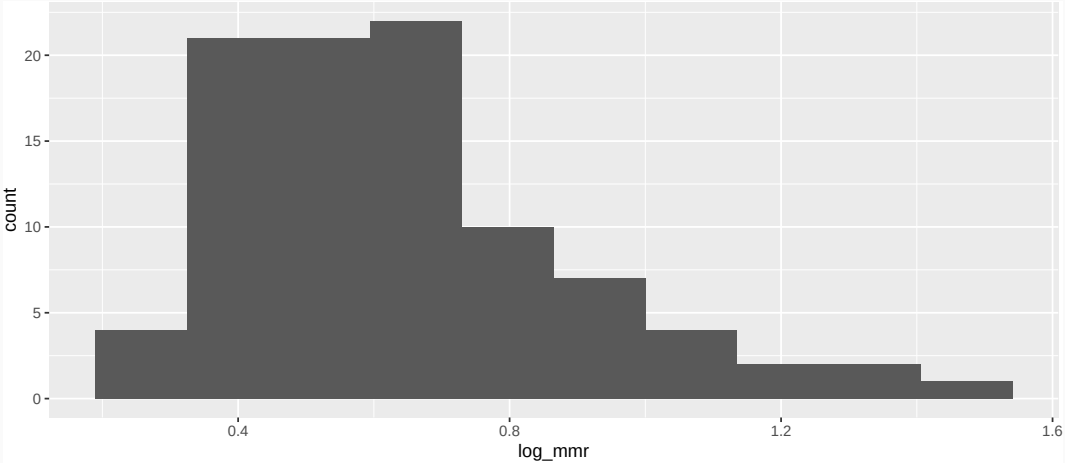
Cannabis data



Cannabis data



Cannabis data



Model specification

$$log(mmr_i) = \beta_0 + \beta_1 \text{pupil}_i + \beta_2 \text{iPad}_i + \epsilon_i$$

Define the likelihood:

- $\epsilon \sim N(0, \sigma_y^2 I_n) \implies y \sim N(X\beta, \sigma_y^2 I_n)$

Define the priors:

- $\beta \sim N(0, \sigma_\beta^2 I_\beta)$
- $\sigma_y^2 \sim Exp(\lambda)$

Define the hyperparameters

- Let $\sigma_\beta^2 = 10,1000$
- Let $\lambda = 1$

Fitting the model

We will use the R package rstanarm

```
blr_mod = stan_glm(log_mmr ~ p_fpc1 + i_time_outside_reticle + use_group,
                   data = cannabis,
                   prior_intercept = normal(0, 10),
                   prior = normal(0, 10),
                   prior_aux = exponential(rate = 1))
```

```
SAMPLING FOR MODEL 'continuous' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 0.000242 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 2.42 s
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
```

Model summary

```r
summary(blr_mod)
```

```
Model Info:
 function:     stan_glm
 family:       gaussian [identity]
 formula:      log_mmr ~ p_fpc1 + i_time_outside_reticle + use_group
 algorithm:    sampling
 sample:       4000 (posterior sample size)
 priors:       see help('prior_summary')
 observations: 94
 predictors:   4

Estimates:
                         mean   sd   10%   50%   90%
(Intercept)              1.0    0.1  0.8   1.0   1.2
p_fpc1                   0.0    0.0  0.0   0.0   0.0
i_time_outside_reticle   0.0    0.0  0.0   0.0   0.0
use_groupdaily use      -0.2    0.1 -0.2  -0.2  -0.1
sigma                    0.2    0.0  0.2   0.2   0.3
```

Fitting the model- diffuse-r priors

This fitting was substantially slower than the previous model.

```
blr_mod2 = stan_glm(log_mmr ~ p_fpc1 + i_time_outside_reticle + use_group,
                    data = cannabis,
                    prior_intercept = normal(0, 1000),
                    prior = normal(0, 1000),
                    prior_aux = exponential(rate = 1))
```

```
SAMPLING FOR MODEL 'continuous' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 1.2e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.12 s
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
```

Model summary

```r
summary(blr_mod2)
```

```
Model Info:
 function:     stan_glm
 family:       gaussian [identity]
 formula:      log_mmr ~ p_fpc1 + i_time_outside_reticle + use_group
 algorithm:    sampling
 sample:       4000 (posterior sample size)
 priors:       see help('prior_summary')
 observations: 94
 predictors:   4

Estimates:
                         mean   sd   10%   50%   90%
(Intercept)              1.0    0.1  0.8   1.0   1.2
p_fpc1                   0.0    0.0  0.0   0.0   0.0
i_time_outside_reticle   0.0    0.0  0.0   0.0   0.0
use_groupdaily use      -0.2    0.1 -0.2  -0.2  -0.1
sigma                    0.2    0.0  0.2   0.2   0.3
```

OLS

```r
##### compare with least squares
mod_lm = lm(log_mmr ~ p_fpc1 + i_time_outside_reticle + use_group, data = cannabis)
summary(mod_lm)
```

```
Call:
lm(formula = log_mmr ~ p_fpc1 + i_time_outside_reticle + use_group,
    data = cannabis)

Residuals:
     Min       1Q   Median       3Q      Max
-0.47538 -0.16440 -0.05425  0.12442  0.90746

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)             0.9921595  0.1298175   7.643 2.21e-11 ***
p_fpc1                 -0.0001831  0.0004189  -0.437  0.66300
i_time_outside_reticle -0.0033046  0.0017121  -1.930  0.05673 .
use_groupdaily use     -0.1651633  0.0551611  -2.994  0.00355 **
---
```

Regression coefficients

term	estimate	std.error	conf.low	conf.high	model
(Intercept)	0.9856	0.1291	0.7801	1.2099	blr
(Intercept)	0.9922	0.1298	0.7343	1.2501	lm
(Intercept)	0.9942	0.1326	0.7779	1.2141	blr_diffuse
i_time_outside_reticle	-0.0032	0.0017	-0.0061	-0.0005	blr
i_time_outside_reticle	-0.0033	0.0017	-0.0067	0.0001	lm
i_time_outside_reticle	-0.0033	0.0018	-0.0062	-0.0004	blr_diffuse
p_fpc1	-0.0002	0.0004	-0.0008	0.0005	blr
p_fpc1	-0.0002	0.0004	-0.0010	0.0006	lm
p_fpc1	-0.0002	0.0004	-0.0009	0.0005	blr_diffuse
use_groupdaily use	-0.1645	0.0555	-0.2578	-0.0743	blr
use_groupdaily use	-0.1652	0.0552	-0.2748	-0.0556	lm
use_groupdaily use	-0.1649	0.0569	-0.2616	-0.0740	blr_diffuse

Bayesian regression diagnostics

The package `rstanarm` uses MCMC to draw samples from the posterior distribution. There are standard checks to ensure that your results come from independent draws of the posterior distribution.

- We will come back to these after we talk more about MCMC
- `stan_glm` has lots of nice tutorials